

University of Asia Pacific
Department of Computer Science and Engineering
REPORT: 08

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1. Problem Statement

Implement the FIFO (First In, First Out) and LRU (Least Recently Used) page replacement algorithms using Python for the following reference string:

Reference String: 1, 2, 3, 4, 2, 3, 1
Number of Page Frames: 3

Calculate the total number of page faults for both algorithms and compare their performance.

2. Objective

The objective of this experiment is to implement FIFO and LRU page replacement algorithms using Python. It also aims to understand how these algorithms replace pages when memory frames are full and compare their total page faults.

3. Introduction

Page replacement is an important memory management technique in an operating system. When all memory frames are occupied and a new page needs to be loaded, the operating system must select an existing page for replacement.
FIFO replaces the page that entered memory first, whereas LRU replaces the page that has not been used for the longest time.

4. Algorithm Description

FIFO Algorithm:

•Check whether the requested page is present in memory.
•If it is absent, count a page fault.
•If memory is full, remove the oldest page and insert the new page.
LRU Algorithm:

•Check whether the requested page is present in memory.
•If it is absent, count a page fault.
•If memory is full, replace the least recently used page.

5. Source Code

The Python program uses lists to represent memory frames and processes each page in the reference string. It counts page faults separately for FIFO and LRU and displays the results for comparison.

6. Result

Reference String: 1, 2, 3, 4, 2, 3, 1
Number of Page Frames: 3
FIFO Page Faults: 5
LRU Page Faults: 5

7. Comparison Between FIFO and LRU

FIFO and LRU are two common page replacement algorithms used in operating systems. FIFO replaces the page that entered memory first, whereas LRU replaces the page that has not been used for the longest time.

FIFO is simple to implement because it follows the arrival order of pages. However, it does not consider how frequently or recently a page is used. LRU considers recent page usage, which can help keep useful pages in memory, but it requires additional tracking.

For the given reference string (1, 2, 3, 4, 2, 3, 1) with three page frames, both algorithms produce the same result.

FIFO Page Faults: 5
LRU Page Faults: 5

Therefore, both algorithms perform equally in this experiment. However, their performance may differ for other reference strings. FIFO is simpler, while LRU may produce fewer page faults when recently used pages are accessed again.

8. Discussion

In this experiment, FIFO and LRU were implemented using Python. Both algorithms produced 5 page faults for the given reference string. Students may face difficulties tracking memory frames and deciding which page should be replaced. FIFO follows the arrival order, whereas LRU considers recent page usage. The experiment demonstrates that different replacement strategies can produce different results depending on the reference string.

9. Conclusion

The experiment successfully implemented FIFO and LRU page replacement algorithms using Python. Both algorithms produced 5 page faults with three memory frames. This experiment helped us understand how page replacement works and how different strategies affect memory management performance. Although LRU may perform better for some reference strings, the best result depends on the page access pattern.

GitHub Repository link :

https://github.com/Rajash144/OS-CPU-Scheduling-Labs